

a) $C_7H_{12}O_4$

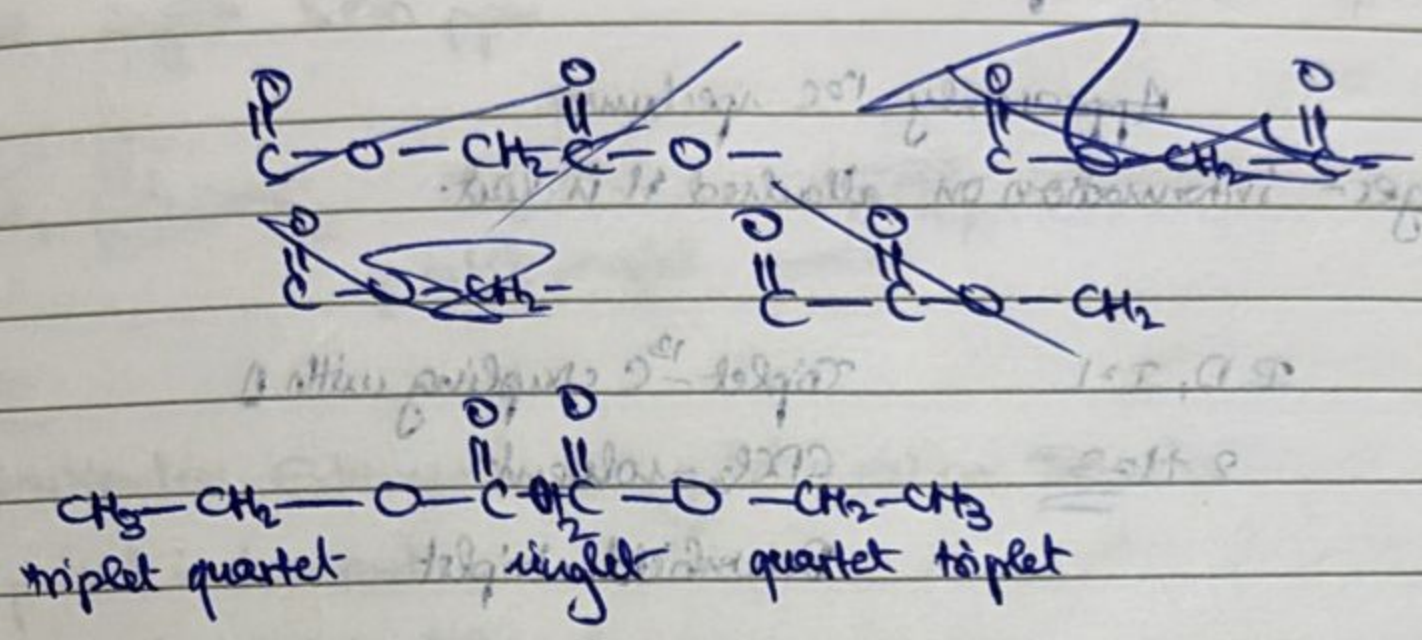
$$IHD = \frac{16 - 12}{2} = 2$$

1734 cm⁻¹ (C=O) → 1700 cm⁻¹

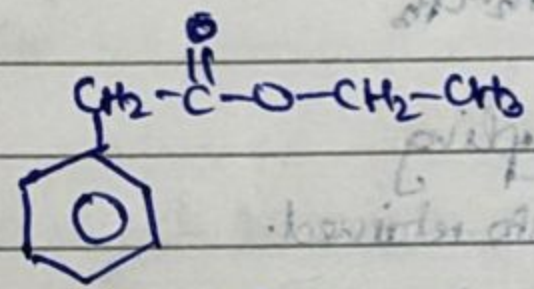
Several (s) centering at around 1200 cm⁻¹ in IR spectrum.

C-O (1200-1000 cm⁻¹)

4O → 2 ester groups



8-4-26 ^{13}C coupled to the hydrogens



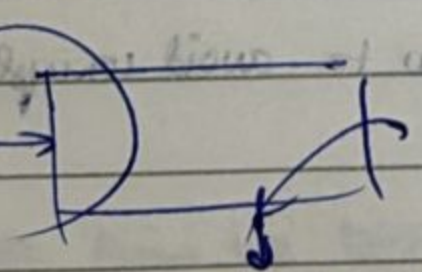
3 diff types of C atoms (o, m, p different)

Proton-Decoupled ^{13}C Spectra

- * Irradiate with a particular RF.
- * Irradiate all of the H-nuclei in the molecule at the same time the carbon resonances are being measured.

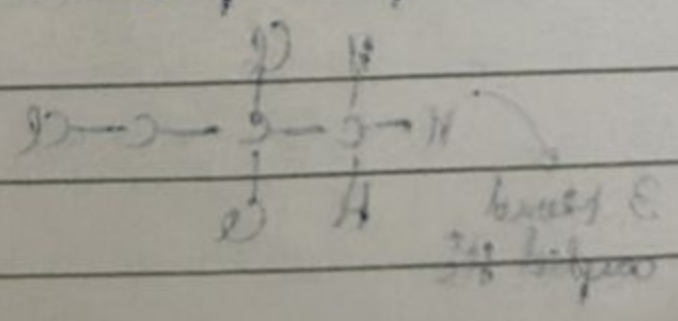
RF source

"the decoupler" continuously saturates hydrogens



RF source 1 pulse tuned to C-13

^{13}C signal measured



sample irradiated with 2 diff. RF

1- to excite all C nuclei

2- to cause all protons rapid transitions

Decoupled any spin-spin interaction b/w ^{13}C and ^1H nuclei

Rapid change
All spin interactions averaged to 0

Appear only ^{13}C spectrum

Disadvantage - information on attached H is lost.

$[2n+1]$

$\text{H}, n = \frac{1}{2}$

$\therefore n+1$

$2D, T=1$

$2H=3$

Triplet - ^{13}C coupling with 3

CDCl_3 - solvent

Downfield - triplet

$\text{OH}-\text{CH}_2-\text{CH}_2-\text{CH}_3$

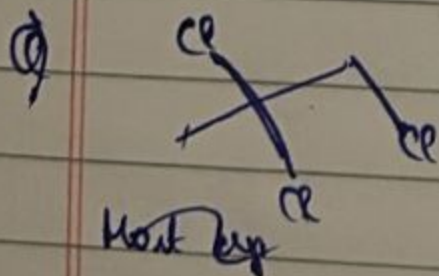
Off-Resonance Decoupling

- * some of splitting info retained.
- * One bond coupling $^{13}\text{C}-^1\text{H}$. Beyond that, $\times$
- * $\times$

- * Decoupler is set either upfield or downfield.

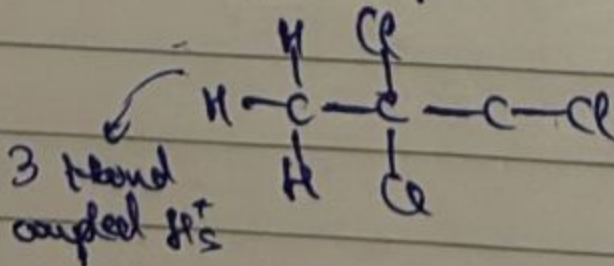
- * Either upfield or downfield from TMS

- * Power of decoupling oscillator low to avoid complete decoupling.



$2D, T=1$

Most upfield peak - CH_3 - multiplicity of 4



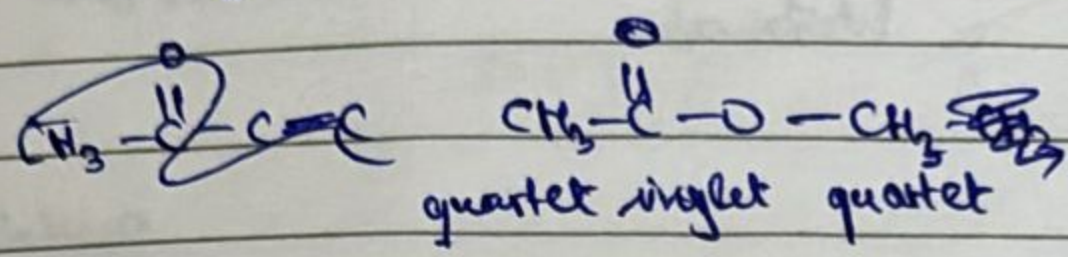
$^{13}\text{C}-^1\text{H}$ coupling
(Off-resonance, only this)

CH₂ triplet
C singlet

Q) CH₂ 9

$$\Delta HD = \frac{8-6}{2} = 1$$

C=O singlet 8.170 ppm



DEPT

- * Distortionless Enhancement by Polarization Transfer
- * FT pulsed instrument
- * ¹³C signals exhibit different phases
- * Duration of the complex pulse
- * Sub spectra
- * Distinguishes ¹³C signals for CH₃, CH₂, CH and quaternary carbons

Broadband-decoupled	DEPT-45	DEPT-90	DEPT-135
C, CH, CH ₂ , CH ₃	All C that have H attached (all except 4°)	CH	CH ₃ , CH are +ve CH ₂ is -ve

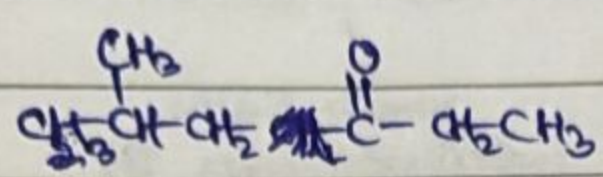
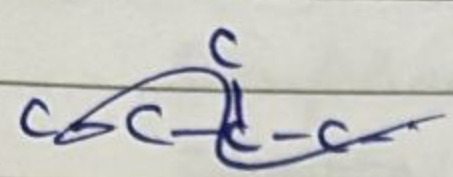
C Subtract DEPT-135 from broadband-decoupled spectrum

CH DEPT-90

CH₂ -ve DEPT-135

CH₃ Subtract DEPT-90 from +ve DEPT-135

Isopentyl acetate



DEPT-90 1
DEPT-135 2 3
DEPT-135 6

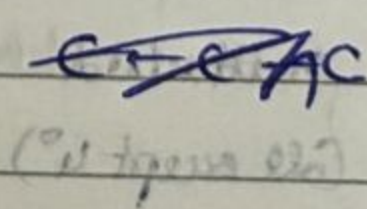
Advantages of ¹³C NMR

- * Info about backbone of the molecule rather than the periphery.
- * Less overlapping of peaks.
- * Homonuclear spin-spin coupling is not observed.
- * Heteronuclear spin-spin coupling b/w ¹³C and ¹²C does not occur.
- * Decoupling b/w ¹³C and ¹H.

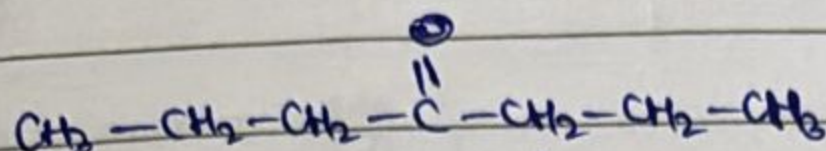
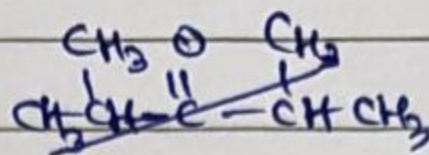
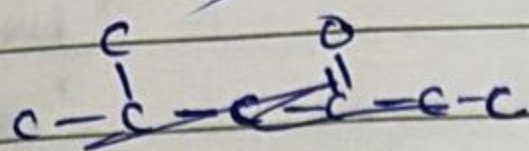
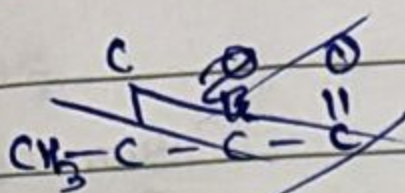
Q) ¹³C NMR

$$\Delta HD = \frac{16-14}{2} = 1$$

¹³C 4 peaks
211.04
44.79
17.39



2 2 3
protons (x2) (14H)
4 4 6

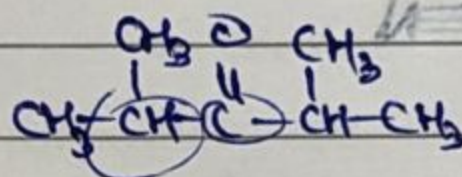
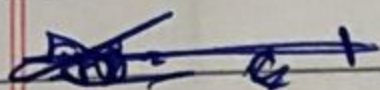


Assign peaks

most downfield

2) $\text{C}_7\text{H}_{14}\text{O}$

Septet

Doublet $\rightarrow$ Propyl3) $\text{C}_7\text{H}_{14}\text{O}$ ^{13}C 5 peaks 28.31

32.27

30.88

29.73

55.98

2

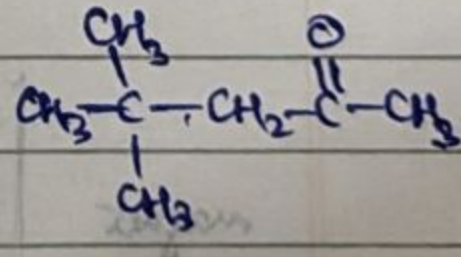
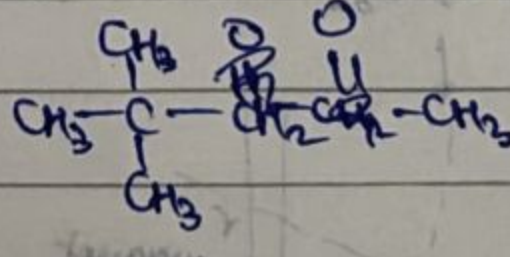
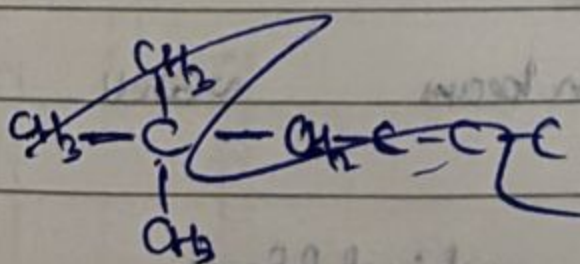
 ^1H

2

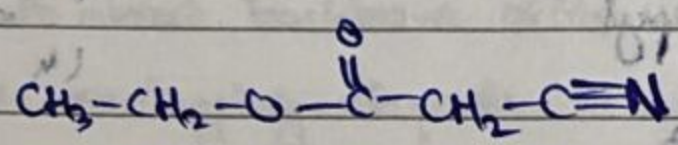
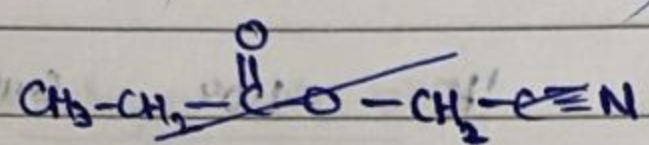
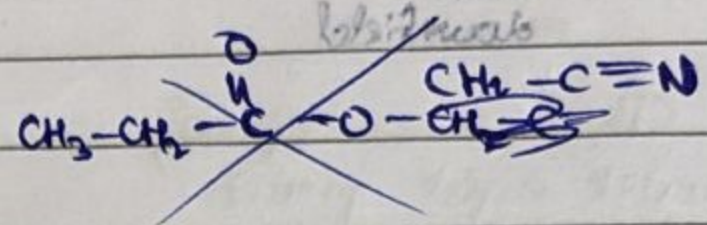
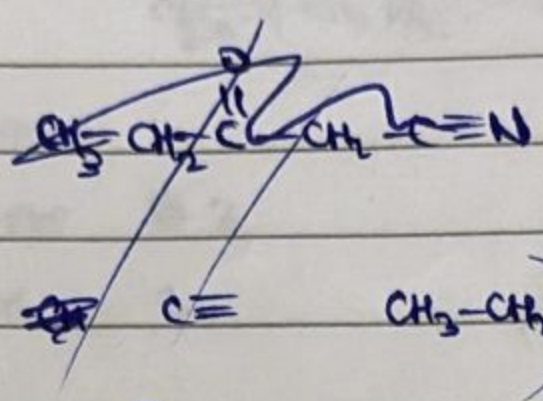
3

(9)

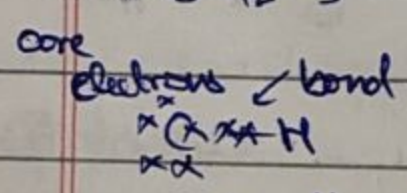
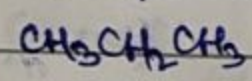
All 3 singlets



Q) $C_5H_7NO_2$
 $IND = \frac{12 + 1 - 7}{2} = \frac{6}{2} = 3$

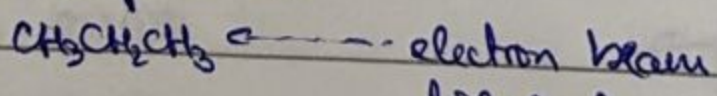


10-426 Mass Spectrometry

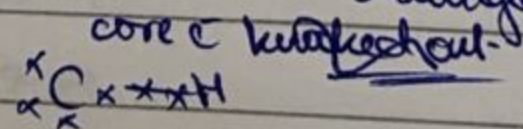
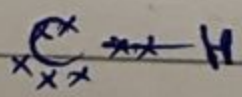


bonding electrons

mass spectrometer

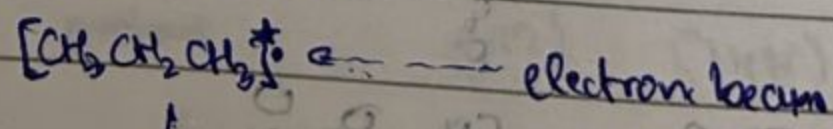


dislodged



molecular ion

radical cation



~60 eV

magnet

magnet

centrifugal force

detector